Selected Attributes Model - Strategy/Logic

Step 1: Removed bottom 5 attributes with the lowest Gini from the Loss Reduction Variable Importance (Random Forest) table

- The five attributes with the lowest Gini index in the Random Forest Loss Reduction Variable Importance table were removed
- Removed:
  - **Last_Loan_9**
  - **Last_Loan_8**
  - **Last_Loan_7**
  - **Payment_Behaviour**
  - **Last_Loan_6**

Loss Reduction Variable Importance					
Variable	Number of Rules	Gini	OOB Gini	Margin	OOB Margin
Credit_Mix	1700	0.059132	0.05881	0.067561	0.06708
Interest_Rate	24638	0.053528	0.03503	0.062063	0.04367
Payment_of_Min_Amount	1160	0.029420	0.02909	0.019769	0.01944
Outstanding_Debt	35036	0.054194	0.02675	0.082332	0.05481
Delay_from_due_date	25849	0.037195	0.01379	0.046394	0.02262
Num_Credit_Inquiries	28977	0.034155	0.01181	0.043710	0.02121
Num_Credit_Card	26156	0.023815	0.00631	0.043428	0.02587
Num_Bank_Accounts	25290	0.023083	0.00586	0.031894	0.01469
Last_Loan_5	2574	0.005211	0.00190	0.007271	0.00452
Last_Loan_6	2040	0.004350	0.00185	0.003657	0.00170
Credit_History_Age	27732	0.028686	0.00182	0.044083	0.01580
Last_Loan_7	1588	0.002124	0.00039	0.002431	0.00098
Last_Loan_9	412	0.000442	0.00002	0.000723	0.00038
Last_Loan_8	788	0.000715	-0.00008	0.001309	0.00067
Changed_Credit_Limit	22389	0.021935	-0.00027	0.040815	0.01802
Last_Loan_4	4675	0.004688	-0.00054	0.006713	0.00257
Last_Loan_2	7359	0.008082	-0.00084	0.012660	0.00560
Num_of_Loan	23044	0.013568	-0.00136	0.021758	0.00687
Last_Loan_3	6213	0.006044	-0.00140	0.008990	0.00306
Num_of_Delayed_Payment	32518	0.026070	-0.00175	0.036610	0.00846
Payment_Behaviour	6361	0.004031	-0.00192	0.006282	0.00139
Last_Loan_1	7745	0.007475	-0.00229	0.011778	0.00409
Annual_Income	18161	0.013821	-0.00252	0.023822	0.00741
Monthly_Inhand_Salary	27114	0.018755	-0.00391	0.031755	0.00905
Age	20559	0.014451	-0.00408	0.025174	0.00611
Occupation	7841	0.009038	-0.00486	0.014594	0.00516
Total_EMI_per_month	33279	0.020064	-0.00532	0.036222	0.01062
Amount_invested_monthly	23216	0.015277	-0.01229	0.028057	-0.00027
Monthly_Balance	31180	0.018532	-0.01512	0.034363	-0.00044
Credit_Utilization_Ratio	27465	0.016206	-0.01573	0.030380	-0.00254

Obs	Variable	Gini
1	Credit_Mix	0.059132
2	Outstanding_Debt	0.054194
3	Interest_Rate	0.053528
4	Delay_from_due_date	0.037195
5	Num_Credit_Inquiries	0.034155
6	Payment_of_Min_Amount	0.029420
7	Credit_History_Age	0.028686
8	Num_of_Delayed_Payment	0.026070
9	Num_Credit_Card	0.023815
10	Num_Bank_Accounts	0.023083
11	Changed_Credit_Limit	0.021935
12	Total_EMI_per_month	0.020064
13	Monthly_Inhand_Salary	0.018755
14	Monthly_Balance	0.018532
15	Credit_Utilization_Ratio	0.016206
16	Amount_invested_monthly	0.015277
17	Age	0.014451
18	Annual_Income	0.013821
19	Num_of_Loan	0.013568
20	Occupation	0.009038
21	Last_Loan_2	0.008082
22	Last_Loan_1	0.007475
23	Last_Loan_3	0.006044
24	Last_Loan_5	0.005211
25	Last_Loan_4	0.004688
26	Last_Loan_6	0.004350
27	Payment_Behaviour	0.004031
28	Last_Loan_7	0.002124
29	Last_Loan_8	0.000715
30	Last_Loan_9	0.000442

Step 2: Remove bottom 2 attributes from Variable Importance (Decision Tree) table

- The two attributes with the lowest Variable Importance scores in the Decision Tree analysis were removed
- Removed:
  - **Credit_Utilization_Ratio**
  - **Monthly_Balance**

Variable Importance			
	Training		
Variable	Relative	Importance	Count
Credit_Mix	1.0000	69.7447	1
Outstanding_Debt	0.7900	55.0998	10
Interest_Rate	0.3832	26.7243	12
Delay_from_due_date	0.2957	20.6259	8
Occupation	0.2668	18.6114	38
Credit_History_Age	0.2644	18.4392	13
Changed_Credit_Limit	0.2408	16.7932	6
Num_Credit_Card	0.2402	16.7551	4
Last_Loan_3	0.2001	13.9568	26
Num_of_Delayed_Payment	0.1977	13.7910	7
Last_Loan_2	0.1778	12.3991	20
Last_Loan_1	0.1728	12.0550	18
Annual_Income	0.1728	12.0548	15
Num_Bank_Accounts	0.1568	10.9377	5
Age	0.1555	10.8437	15
Num_Credit_Inquiries	0.1488	10.3792	5
Last_Loan_4	0.1292	9.0107	10
Total_EMI_per_month	0.1254	8.7454	8
Last_Loan_5	0.1172	8.1708	7
Amount_invested_monthly	0.1053	7.3420	4
Payment_Behaviour	0.0997	6.9543	8
Monthly_Inhand_Salary	0.0889	6.2034	6
Last_Loan_7	0.0791	5.5151	3
Last_Loan_9	0.0703	4.9045	2
Last_Loan_8	0.0701	4.8875	2
Last_Loan_6	0.0536	3.7402	2
Monthly_Balance	0.0387	2.7015	1
Credit_Utilization_Ratio	0.0271	1.8879	1

Step 3: Remove 2 attributes exhibiting weakest correlation with class attribute
(Credit_Utilization_Ratio already removed in Step 2)

> Correlation strength was evaluated between the target variable (Credit_Score) and numerical attributes (using Eta Squared) as well as categorical attributes (using Cramér's V). Based on this analysis, two attributes were removed:

	Variable	P_Value	Significance	Cramers_V	Association_Strength
4	Occupation	<.0001	Significant	0.03005710	Weak association
13	Last_Loan_9	<.0001	Significant	0.08719994	Weak association
3	Payment_Behaviour	<.0001	Significant	0.09071711	Weak association
5	Last_Loan_1	<.0001	Significant	0.11316613	Moderate association
12	Last_Loan_8	<.0001	Significant	0.12228283	Moderate association
6	Last_Loan_2	<.0001	Significant	0.16664049	Moderate association
7	Last_Loan_3	<.0001	Significant	0.17406019	Moderate association
11	Last_Loan_7	<.0001	Significant	0.17677884	Moderate association
8	Last_Loan_4	<.0001	Significant	0.21025318	Moderate association
10	Last_Loan_6	<.0001	Significant	0.22503579	Moderate association
9	Last_Loan_5	<.0001	Significant	0.27128322	Moderate association
1	Credit_Mix	<.0001	Significant	0.44994121	Strong association
2	Payment_of_Min_Amount	<.0001	Significant	0.47350231	Strong association

	Numeric_Variable	P_Value	Eta_Squared	Significance	Interpretation
13	Credit_Utilization_Ratio	<.0001	0.002101572	Significant	Weak correlation
15	Total_EMI_per_month	<.0001	0.005766290	Significant	Weak correlation
16	Amount_invested_monthly	<.0001	0.024671499	Significant	Weak correlation
1	Age	<.0001	0.025983541	Significant	Weak correlation
17	Monthly_Balance	<.0001	0.043503224	Significant	Weak correlation
3	Monthly_Inhand_Salary	<.0001	0.044612069	Significant	Weak correlation
2	Annual_Income	<.0001	0.045799318	Significant	Weak correlation
10	Changed_Credit_Limit	<.0001	0.050934712	Significant	Weak correlation
7	Num_of_Loan	<.0001	0.129539411	Significant	Moderate correlation
9	Num_of_Delayed_Payment	<.0001	0.150246651	Significant	Strong correlation
14	Credit_History_Age	<.0001	0.151161105	Significant	Strong correlation
12	Outstanding_Debt	<.0001	0.154202003	Significant	Strong correlation
4	Num_Bank_Accounts	<.0001	0.156428677	Significant	Strong correlation
5	Num_Credit_Card	<.0001	0.163584909	Significant	Strong correlation
8	Delay_from_due_date	<.0001	0.186624633	Significant	Strong correlation
11	Num_Credit_Inquiries	<.0001	0.192492734	Significant	Strong correlation
6	Interest_Rate	<.0001	0.235716366	Significant	Strong correlation

- Removed:
  - **Total_EMI_per_month**
  - **Amount_invested_monthyl**

Step 4: Remove attributes not present in Variable Importance (Decision Tree) table & also in bottom half of Loss Reduction Variable Importance (Random Forest) table

- Attributes that were not present in the Decision Tree Variable Importance table and ranked in the bottom half of the Random Forest Loss Reduction Variable Importance table were removed
- Removed:
  - **Num_of_Loan**

Obs	Variable	Gini
1	Credit_Mix	0.059132
2	Outstanding_Debt	0.054194
3	Interest_Rate	0.053528
4	Delay_from_due_date	0.037195
5	Num_Credit_Inquiries	0.034155
6	Payment_of_Min_Amount	0.029420
7	Credit_History_Age	0.028686
8	Num_of_Delayed_Payment	0.026070
9	Num_Credit_Card	0.023815
10	Num_Bank_Accounts	0.023083
11	Changed_Credit_Limit	0.021935
12	Total_EMI_per_month	0.020064
13	Monthly_Inhand_Salary	0.018755
14	Monthly_Balance	0.018532
15	Credit_Utilization_Ratio	0.016206
16	Amount_invested_monthly	0.015277
17	Age	0.014451
18	Annual_Income	0.013821
19	Num_of_Loan	0.013568
20	Occupation	0.009038
21	Last_Loan_2	0.008082
22	Last_Loan_1	0.007475
23	Last_Loan_3	0.006044
24	Last_Loan_5	0.005211
25	Last_Loan_4	0.004688
26	Last_Loan_6	0.004350
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Occupation	7841	0.009038	-0.00486	0.014594	0.00516
Total_EMI_per_month	33279	0.020064	-0.00532	0.036222	0.01062
Amount_invested_monthly	23216	0.015277	-0.01229	0.028057	-0.00027
Monthly_Balance	31180	0.018532	-0.01512	0.034363	-0.00044
Credit_Utilization_Ratio	27465	0.016206	-0.01573	0.030380	-0.00254